

VOLUME IV - REFERENCES / GOVERNMENT REPORTS

Government Reports and Regulatory Instruments

Statutory, Policy and Institutional Context for the Programme

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ABSTRACT

An index of the statutory instruments, regulatory circulars, national policy documents and institutional sources that define the envelope within which any Bailadila low-visibility haulage system must operate. Each entry records what the instrument requires, how it bears on the proposed system, and what action it implies for the programme. The document closes with a compliance obligation matrix and a note on the regulatory approval pathway for functions that take control authority.

Prepared as part of a solution-design research package addressing dense-fog and low-visibility haul road operations in large-scale mechanised open cast iron ore mines. Figures relating to NMDC production, statutory provisions and commercial systems are compiled from public sources; site-specific values marked as indicative require validation against NMDC operational records before use in a formal proposal.

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1. Scope and Caveat

This index identifies the instruments relevant to the programme and summarises their bearing on it. It is a research aid, not legal advice, and it is not a substitute for reading the instruments themselves.

Currency caveat. Statutory instruments, circulars and policy documents are amended. Every provision summarised here must be verified against the current published text before it is relied upon in a specification, tender or approval submission. Where this package states a requirement as deriving from a circular, the programme should obtain and cite the circular directly.

2. Primary Legislation

GR-A1. The Mines Act, 1952. The governing statute for mine safety in India, establishing the Directorate General of Mines Safety, the duties of owners, agents and managers, the inspectorate's powers, and the framework for regulations made under it. *Bearing on the programme:* responsibility under the Act attaches to both the lease holder and any contractor deploying HEMM on the lease - relevant because Bailadila operations involve contractor fleets, and because a collision avoidance system's coverage must extend to contractor machines for the cooperative layer to be meaningful. *Action:* confirm contractual mechanisms obliging contractors to accept system fitment.

GR-A2. The Metalliferous Mines Regulations, 1961 (MMR 1961). The principal regulations for non-coal mines, including iron ore. Cover haul road construction and maintenance, transport machinery, the duties of machinery operators, illumination, signage, and the general conduct of surface transport. *Bearing:* Bailadila is a metalliferous operation and MMR 1961 rather than the Coal Mines Regulations is the governing instrument. Requirements on haul road width, gradient, berms, signals and transport discipline form the baseline that this programme supplements and must not contradict. *Action:* map every proposed procedural change - particularly the graded operating classes V0 to V4 - against the relevant MMR provisions, and identify where a change requires permission or relaxation.

GR-A3. Regulations 107 and 108 (surface transport and machinery provisions). Cited in DGMS material as the regulatory basis, with Circular 08/2017, for the requirement that surface transport machinery and HEMM carry audio-visual alarms activating automatically on start-up and reversing. *Bearing:* establishes the existing mandatory warning baseline that the proposed system augments rather than replaces. *Action:* retain existing AVA function; ensure new alert tones are distinguishable from it per requirement HR-04.

GR-A4. The Mines Rules, 1955 and associated subordinate instruments. Cover, among other matters, medical examination, training and welfare. *Bearing:* operator training on a new safety-critical system falls within the training framework; vocational training obligations apply to the induction required for the proposed interface.

3. DGMS Technical Circulars

GR-B1. DGMS Technical Circular No. 03 of 2024 - Safety in Transport and Trimming Machinery (TTM) operations at opencast mines. Issued in response to a cluster of vehicle-related accidents at Indian mining sites. Reported provisions include: mandatory daily pre-operational checklists for all TTM - dumpers, dozers, loaders and graders - signed and maintained; **rear-view cameras and Proximity Detection Systems required on all dumpers**; certified operators with regular refresher training; haul road gradient limited to a maximum of 1 in 16; requirements on haul road width, berms, drainage and signage with potholes repaired regularly; mandatory seat belt use, displayed speed limits, no overloading, spotters or signallers for reversing; collision avoidance systems, reversing alarms and auto-dip headlights; parking discipline including wheel chocks and a prohibition on unattended vehicles on slopes; and preventive maintenance with breakdowns recorded in logbooks. Fatal and serious accidents require notification to DGMS within a short specified period. *Bearing:* **the single most directly relevant instrument.** It already mandates PDS and collision avoidance, which means the proposed system extends an existing obligation rather than introducing a novel one - a significant advantage for approval and for internal justification. The 1 in 16 gradient limit is the input to the stopping-distance derivation in Volume I Document 3, Section 3.1. The pre-operational checklist requirement is the vehicle for the system pre-shift verification of requirement RR-03. *Action:* obtain the circular text; map each provision to a system requirement; integrate system self-test into the existing checklist rather than creating

a parallel one.

GR-B2. DGMS Technical Circular No. 06 of 2020 - Safety features for HEMM and heavy/light vehicles in opencast mines. Reported provisions include: rear vision cameras; operator fatigue warning systems; auto-dipping systems with filtering to prevent false triggering from other light sources such as pole lights; a mechanical device to avoid head-to-tail collision, absorbing or diverting collision energy, to be part of the original design in new dumpers and not to obstruct visibility or affect stability; automatic fire detection and suppression covering hot zones such as engine and exhaust, with both automatic and manual activation and visual and audio alerts; dump body raised position indicator with visual and audio warning and transmission restricted beyond first gear until the body is fully lowered; joystick auto-return to neutral halting tramming on release; and a **proximity warning device detecting static and moving objects including people around the dumper or tipper during movement, providing audio-visual warnings, detecting objects within a virtual target area and adjusting the warning based on speed and direction.** *Bearing:* the proximity warning provision is notable for already requiring speed- and direction-dependent warning adjustment - which is the trajectory-aware principle this package advocates in requirement FR-19, though deployed products commonly implement it only as a short-range envelope. The dump body indicator and transmission interlock are existing precedents for machine-state interlocks, useful when arguing the approval case for any future intervention function. *Action:* document how the proposed system satisfies and exceeds each proximity warning provision.

GR-B3. DGMS Circular 08/2017 - Audio-visual alarms. With Regulations 107/108, requires audio-visual alarms on surface transport and heavy earth moving machines, activating automatically on start-up and reversing. DGMS material describes expectations including an audible alarm producing a distinct sound in the region of 90-110 dB and a visual beacon visible at a substantial distance, with alarm function checked every shift and machines with defective alarms withdrawn from use. *Bearing:* establishes the existing warning baseline and the per-shift verification culture into which system self-test integrates. Also illustrates the limitation this package identifies in Volume I Document 2, Section 2.4 - the audible alarm is omnidirectional and non-informative, and the visual beacon is optical and therefore attenuated in fog.

GR-B4. Other DGMS circulars on haul road design, illumination, and mine transport safety. *Action:* conduct a systematic review of current DGMS circulars applicable to metalliferous opencast transport at programme inception, as part of the regulatory baseline. The review output should be a living compliance register maintained through the programme.

4. National Policy Context

GR-C1. National Steel Policy 2017. Targets an installed domestic crude steelmaking capacity of 300 MT by 2030-31. Associated projections indicate a requirement of approximately **430 to 437 MT of iron ore annually** by that date. *Bearing:* this is the origin of NMDC's 100 MT capacity target and therefore of the Bailadila sector's approximately 80 MT expectation. It frames the programme not as a discretionary safety improvement but as an enabler of a national policy objective - the argument developed in Volume I Document 1, Section 1.3 and Volume II Document 3, Section 9.

GR-C2. Ministry of Steel statements on NMDC performance. Public statements record NMDC's FY 2025-26 production of **53.15 MT** with approximately 21 per cent year-on-year growth and sales of **50.23 MT** with approximately 13 per cent growth, described as the best volumes since inception, and attribute the growth to best-ever performance at Kirandul, Bacheli and Donimalai. The same period is described as a year of expansion including an international office in Dubai, operationalisation of a first coal mine in Jharkhand, and **commissioning of Deposit 4 at Bailadila.** *Bearing:* establishes the production baseline used throughout this package and confirms continuing Bailadila capacity addition against which monsoon constraint must be assessed.

GR-C3. Mineral policy and mine-planning instruments - the Mines and Minerals (Development and Regulation) Act and associated rules, Indian Bureau of Mines approved mining plans, and the star-rating framework under which NMDC mines are rated. *Bearing:* haul road development and any fixed infrastructure siting must be consistent with the approved mining plan; this is directly relevant to constraint C-13 on road mutability and to requirement ER-08 on relocatable roadside nodes.

5. NMDC Corporate and Operational Sources

GR-D1. NMDC Limited corporate disclosures and operating-mine descriptions. Establish the complex and deposit structure relied on in this package: Kirandul Complex comprising Deposit-14 (commissioned April 1968, the first large-scale mechanised open cast iron ore mine in India, with the DPR prepared by NMDC), the 11C Project (14 NMZ + 11C part, commissioned June 1988) and Deposit-11B (commissioned 2015, the most recently developed of the three at that time), sharing common infrastructure with a township at Kirandul; Bacheli Complex working Deposits 5, 10 and 11A; and Donimalai and Kumaraswamy in Karnataka. Also record Bailadila Deposit-13 lease transfer in January 2017 and Kumaraswamy plant operation from June 2017. *Bearing:* the physical scope of the programme.

GR-D2. NMDC expansion and capacity reporting. Public reporting indicates a target of 60 MT in the current fiscal year, described as approximately 20 per cent above the prior year, building on crossing the 50 MT milestone; reliance on increased output from existing mines together with capacity from the NMDC-CMDC Limited joint venture; environmental clearances sought for selected deposits; ongoing construction of Screening Plant III at Kirandul; a slurry pipeline project with a segment from the Bailadila/Bacheli area to Nagarnar; a slime beneficiation plant planned at Kirandul to recover iron ore from tailings; and long-term capital expenditure guidance in the range of Rs 65,000-80,000 crore over a multi-year period toward the 100 MTPA target. Reporting also notes incremental loading capacity from new lines at Bacheli and Kirandul. *Bearing:* confirms that capacity additions address annual average output and do not by themselves address the seasonal distribution problem this package identifies.

GR-D3. Reporting on Bailadila evacuation and operational constraints. Analyst and trade reporting identifies the Bailadila sector as contributing approximately **70 per cent of NMDC's total production** while depending on the single-track **Kirandul-Kothavalasa (KK) railway line**, historically subject to monsoon-related landslides through the Ananthagiri Ghats and to security incidents in the region, with a historical dispatch limit on the line cited as the primary operational ceiling on expansion from a landlocked sector. *Bearing:* important context and an important qualification. **Monsoon haulage constraint and monsoon evacuation constraint are distinct problems.** This package addresses the former. If rail evacuation is the binding constraint during the same months, the production benefit of recovered haulage hours may be partly absorbed by an inability to despatch. *Action:* the Phase 0 baseline (Volume II Document 3) must establish whether, during monsoon months, the binding constraint at Bailadila is mine-side haulage, rail evacuation, or both - because that determines the realisable benefit. This is a material qualification on the business case and should be resolved early.

GR-D4. NMDC sustainability and safety disclosures. Record five-star IBM ratings across NMDC mines, plantation programmes, and a reduction in greenhouse gas emission intensity over a nine-year period from approximately 1311.5 to 999.6 tCO₂e/MT between 2011-12 and 2020-21. *Bearing:* context; also relevant in that reduced stop-start cycling and reduced idling during fog events carry a modest emissions co-benefit worth recording in the benefit model.

6. Spectrum and Telecommunications

GR-E1. Wireless Planning and Coordination (WPC) Wing, Ministry of Communications - spectrum allocation and licensing. *Bearing:* governs authorisation for 76-81 GHz radar, 5.9 GHz ITS band use for C-V2X, sub-GHz bands used by proprietary collision avoidance bearers, UWB, and any captive private LTE/5G spectrum. *Action:* **this is a long-lead item and programme risk R9.** Confirm the position on each candidate band in Phase 0, before bearer selection is finalised. Requirement CR-07 and GR-04 depend on it.

GR-E2. Equipment type approval requirements. All radio equipment requires applicable Indian type approval. *Action:* procurement must verify that a candidate vendor **holds** current approval, not merely that approval is obtainable.

GR-E3. Provisions and guidance relevant to private captive networks in India. *Action:* confirm current position for the V2I backhaul tier.

7. Environmental and Clearance Instruments

GR-F1. Environmental clearances under the Environment (Protection) Act framework, granted by the Ministry of Environment, Forest and Climate Change. Public reporting indicates NMDC has sought clearances for selected deposits in support of the production ramp. *Bearing:* any fixed infrastructure - RTK base stations, repeaters, roadside visibility nodes - sited within the lease must be consistent with the approved clearance conditions and mining plan. Roadside solar nodes are low-impact but not automatically outside the clearance framework. *Action:* confirm with the site environment function before siting fixed assets.

GR-F2. Forest clearance provisions. *Bearing:* relevant where any infrastructure is sited on forest land, which is a material consideration in the Bailadila region.

8. Meteorological Sources

GR-G1. India Meteorological Department (IMD) climatological data for the Bastar plateau and Dantewada district.

Bearing: the source for regional rainfall, humidity and monsoon onset and withdrawal climatology summarised indicatively in Volume I Document 1, Section 3.1. *Action:* **obtain the actual station data rather than relying on the indicative bands in this package.** The rainfall and humidity figures given in Volume I are regional approximations offered to frame the problem, and the programme should replace them with IMD records and, better, with NMDC's own site observations.

GR-G2. IMD definitions and practice for visibility reporting and fog classification. *Bearing:* aligns the package's MOR-based visibility classes with national meteorological convention, which matters when the system's visibility measurements are compared against official observations or used in any external reporting.

GR-G3. IMD nowcasting products and any local automatic weather station coverage. *Bearing:* potential supplementary input to the nowcasting model of requirement FR-04, though the package's assessment is that site-local sensing will dominate skill at the 15-30 minute horizon.

9. Institutional Bodies

Body	Relevance to the programme
Directorate General of Mines Safety (DGMS)	The statutory authority. Approval counterpart for procedural change (GR-05) and for any control-authority function (GR-03). Engage from Phase 0.
Ministry of Steel	NMDC's administrative ministry; owner of the 100 MT trajectory.
Indian Bureau of Mines (IBM)	Mining plan approval; mine star-rating framework.
Ministry of Environment, Forest and Climate Change	Environmental and forest clearances.
WPC Wing, Ministry of Communications	Spectrum authorisation and equipment approval.
India Meteorological Department	Climatological and observational data.
Bureau of Indian Standards	Standards for equipment, ingress protection, electrical safety - see the Standards index, NMDC-LVH-RF-03.
National Institute of Miners' Health / research institutions	Potential partners for occupational and human-factors evaluation.
CSIR laboratories (e.g. CIMFR)	Potential partners for site instrumentation, testing and independent validation per AR-10.

10. Compliance Obligation Matrix

Obligation	Source	System requirement discharging it	Evidence required
Proximity Detection System on all dumpers	GR-B1	FR-11 to FR-13, FR-18 to FR-20	Fitment record; functional test per machine
Rear-view camera on all dumpers	GR-B1	Camera layer, retained	Fitment record
Proximity warning detecting static and moving objects, speed- and direction-adjusted	GR-B2	FR-11, FR-14, FR-19, FR-24	Functional demonstration against the circular's provisions

Obligation	Source	System requirement discharging it	Evidence required
Audio-visual alarm on start-up and reversing	GR-A3, GR-B3	Existing AVA retained; new alerts distinguishable per HR-04	Per-shift check record
Daily pre-operational checklist for all TTM	GR-B1	RR-03 - system self-test integrated into the existing checklist	Signed checklist with system status field
Haul road gradient not exceeding 1 in 16	GR-B1	Input to PR-01 derivation; road model records gradient	Road model records; survey
Certified operators with refresher training	GR-B1	Training programme for the new interface	Training records
Accident notification within the specified period	GR-B1	OR-03, GR-06 event log supporting investigation	Log extract capability demonstrated
Radio equipment type approval and spectrum authorisation	GR-E1, GR-E2	CR-07, GR-04	Approval certificates held by vendor
EMC and non-interference with blast initiation	GR-A2 general duty; site practice	ER-09	Site coexistence test report, countersigned by the blasting authority
Consistency with approved mining plan and clearances	GR-C3, GR-F1	ER-08 relocatable nodes; siting review	Siting approval record
Procedural change where operating practice is altered	GR-A2, GR-05	Revised SOP for V2/V3/V4 operation	Risk assessment and DGMS agreement

11. The Approval Pathway for Control Authority

The most consequential regulatory question in the programme is the treatment of functions that take control authority over a machine - advisory speed enforcement, automatic braking, or any degree of autonomy.

The package's position, stated in principle P7 and requirement GR-03: advisory and intervention functions must be **architecturally and contractually separated**, so that the deployment of advisory capability - which carries the majority of the benefit - is never gated by the approval of intervention capability.

Why this matters. Advisory functions sit comfortably inside the existing regulatory frame. DGMS Circulars 06/2020 and 03/2024 already mandate proximity warning and collision avoidance; a system that warns better, further and in conditions where existing systems cannot is an enhancement of an existing obligation, not a new category. Intervention is a different category: it alters the machine's response to operator input, engages the precedents around interlocks such as the dump-body transmission restriction in GR-B2, and requires a testing and permissioning discussion with DGMS for which there is limited Indian precedent in this specific application.

Recommended approach.

- ? Open the DGMS dialogue in **Phase 0**, before any procurement, presenting the programme as an extension of existing circular obligations.
- ? Seek early informal guidance on the evidence DGMS would expect for a graded visibility-based operating regime (requirement GR-05), since this is a procedural change rather than an equipment change and may be the more straightforward conversation.
- ? Defer the intervention approval discussion to Phase 4, but **initiate it early enough that its lead time does not become the critical path** if NMDC later decides to pursue it.
- ? Offer the resulting MOR-referenced performance dataset (the gap identified as G5) to DGMS as a contribution to the evidence base. It would be the first such dataset in Indian mining and is likely to be of

direct regulatory interest.